

PHYSIOL 578: HW1

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- 1 Describe a physiologic homeostatic system. Do not use an example from class. If you work with others, your answer should be unique (each student should choose a unique homeostatic system). A drawing may be very helpful. Identify the key components of the homeostatic system and describe briefly how they function and interact. Use a single page for your response.

- 2 Check the hummingbird folder in this section. Explain how a hummingbird can fly across the Gulf of Mexico.
- 3 Ohm's Law: explain how ion channels affect input resistance.
- 4 The Nernst Equation: calculate the reversal potential for a channel permeable to a single ion. Name the ion and the concentrations you chose and why you chose them ("because I found them in a book" is not a good answer). Be sure to list all parameters.

Nernst Equation is given by

$$E_{rev} = \frac{RT}{zF} \ln \frac{[\text{Ion}]_{out}}{[\text{Ion}]_{in}}$$

where:

R is the universal ideal gas constant

T is temperature in kelvins

z is the charge

F is Faraday's constant

- 5 GHK equation: use physiologic concentrations appropriate for the squid giant axon. Estimate the resting potential of the axon. What might complicate your answer? Be sure to list all parameters you used in your calculation.